Theory note S3b — Jamming, a fade or a hop collision? what the pilot residual shows beyond its energy

Chang Young Ham

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Purpose. The S3a energy test notices that a hop is disturbed but not why. S3b compares three causes that all raise the energy of the pilot residual, asks how well they can be told apart from the same $n = 16$ residual samples, and checks the classifiers against two bounds that follow from the model. Mathematics used: Gaussian and chi-square distributions, the Neyman–Pearson lemma, logistic regression.

1 Three causes with the same mean energy

As in S3a, each hop carries n known BPSK pilots $s_i = \pm 1$ (amplitude $\sqrt{E_b} = 1$), the thermal noise has variance $\sigma^2 = N_0/2$, and the receiver subtracts the pilots at their nominal amplitude, leaving the residual $x_i = r_i - s_i$. With $w_i \sim \mathcal{N}(0, \sigma^2)$:

$$\begin{aligned} \text{clean:} \quad & x_i = w_i, \\ \text{jammed:} \quad & x_i = \sqrt{1+D}\, w_i, \\ \text{faded:} \quad & x_i = -g\, s_i + w_i, \qquad g = \sqrt{D}\,\sigma, \\ \text{collision:} \quad & x_i = c\, b_i + w_i, \qquad c = \sqrt{D}\,\sigma,\ b_i = \pm 1 \text{ at random.} \end{aligned}$$

The jammed hop is the S3a model with noise ratio $a = 1 + D$. In the faded hop the signal gain has dropped from 1 to $1 - g$ and the receiver has not tracked it, so part of each pilot stays in the residual. In the collision another node sends its own BPSK symbols b_i on the same channel. D, the disturbance-to-noise ratio, is set equal for the three causes, so each has the same mean residual energy $(1 + D)\sigma^2$ per sample. With $\sigma^2 = 0.05$ the grid $D \le 12\,\text{dB}$ keeps the faded gain $1 - g$ positive.

Let $T = \sum_i x_i^2/\sigma^2$. A faded and a collided hop give exactly the same distribution of T: noncentral chi-square with n degrees of freedom and noncentrality nD, because $s_i^2 = b_i^2 = 1$. A jammed hop gives $T/(1+D) \sim \chi^2_n$, with the same mean but a wider spread (variance $2n(1+D)^2$ against $2n(1+2D)$). So a fade and a collision can only be told apart by the shape of the residual, never by its energy.

2 Two shape features and their means

Besides T, the second classifier uses

$$C = \frac{1}{\sigma\sqrt{n}} \sum_{i=1}^{n} x_i s_i, \qquad K = n\, \frac{\sum_i x_i^4}{\left(\sum_i x_i^2\right)^2}. \tag{1}$$

C measures how much of the pilot pattern is left in the residual. For a faded hop $C = -g\sqrt{n}/\sigma + Z$ with $Z \sim \mathcal{N}(0,1)$, so $\mathbb{E}[C] = -\sqrt{nD}$; for the other three classes $\mathbb{E}[C] = 0$.

K is the sample kurtosis. If the residual is Gaussian with zero mean (clean or jammed hop), $u = x/\|x\|$ is uniformly distributed on the unit sphere, and $\mathbb{E}[u_i^4] = 3/(n(n+2))$, which gives

$$\mathbb{E}[K] = \frac{3n}{n+2} = 2.667 \quad (n = 16) \tag{2}$$

at any noise level. A collision pushes the residual towards the two values $\pm c$ and lowers K towards 1; a fade does the same, but with the sign fixed by the pilot, which C detects.

`VERIFICATION 1` generates 50,000 hops per class at $D = 0$ and 6 dB, compares the sample means of T, C and K with these values ($|z| < 4$, as in S0; largest $|z| = 2.60$ over 20 means), and tests the T of faded and collided hops against the noncentral chi-square distribution (Kolmogorov–Smirnov, $p > 10^{-4}$; smallest $p = 0.38$ over four tests). This checks the data generator before any classifier is trained on it.

3 Three classifiers

Each is trained on 20,000 hops per class at each D and tested on another 20,000 per class (four classes, equal priors, chance 0.25). All three use standard library implementations [1].

- *Energy only*: multinomial logistic regression on $\log T$ and $(\log T)^2$. The square is needed because jammed hops have the widest spread of T and occupy both its low and its high end; a rule linear in $\log T$ can give each class only one interval.
- *Three features*: multinomial logistic regression on $(\log T, C, K)$.
- *Small MLP*: two hidden layers of 32 units on the 16 sorted values $y_i = x_i s_i/\sigma$. Their sum is $\sqrt{n}\,C$, so the pilot correlation is available to it; T and K are not given. Sorting removes the pilot order, which carries no information in this model.

4 Two bounds used as checks

Best accuracy from the energy (`VERIFICATION 2`). With equal priors, the best possible four-class accuracy from T alone is $\frac{1}{4}\int \max_k f_k(t)\,dt$, where f_k are the four densities of Section 1. Because the faded and collided densities are identical, at most one of them can be chosen at any t, so this bound is always below $\frac{1}{4}(1 + 1 + \frac{1}{2} + \frac{1}{2}) = 0.75$. The script integrates it numerically at each D; the energy-only classifier must not exceed it by more than four standard errors. It does not exceed it at any D and stays within 0.010 of it.

Neyman–Pearson bound for clean against jammed hops (`VERIFICATION 3`). S3a showed that for these two classes the likelihood ratio increases with T, so the energy test $T > \gamma$ is the most powerful test: $P_\mathrm{d} = P(\chi^2_n > \gamma/a)$ at $P_\mathrm{fa} = P(\chi^2_n > \gamma)$. A classifier score used with a threshold is also a test, so no classifier can lie above this ROC curve; a point above it would mean an error in the generator or in the scoring.

The check uses the classifiers trained at $D = 0$ dB ($a = 2$). Thresholds for $P_\mathrm{fa} = 0.1$ and 0.01 are set on one set of 200,000 clean hops, and P_fa and P_d are measured on two further independent sets. The tolerance is four standard deviations and includes the error of the measured P_fa, multiplied by the slope of the curve at that point (0.005 at $P_\mathrm{fa} = 0.1$, 0.011 at 0.01). The energy score must lie on the curve (two-sided); the two classifiers must not be above it (one-sided). All six comparisons pass. At $P_\mathrm{fa} \approx 0.01$ the bound is about $P_\mathrm{d} = 0.449$; the three-feature classifier gives 0.452 and the MLP 0.446, so neither is significantly above it and both are within 0.004 of it at the two tested points.

5 What the numbers show

From `results/s3b_summary.txt`:

- The energy-only classifier reaches the energy bound: 0.43 at $D = 0\,\mathrm{dB}$, 0.55 at 6 dB and 0.61 at 12 dB (bound 0.62). Its errors are among the three disturbed classes, and faded and collided hops receive the same mix of labels, as they must.
- With C and K the accuracy rises to 0.69 at 0 dB, 0.92 at 6 dB and 0.997 at 12 dB.
- A fade is recognised almost always (0.965 at 0 dB), because C alone separates it. The remaining errors are between jamming and collision, which differ only in shape: at 0 dB the three-feature classifier labels 42 % of jammed hops as collisions, and the MLP 39 %.
- The MLP matches the three-feature regression up to about 2 dB and does slightly better above (0.937 against 0.922 at 6 dB, mostly fewer jammed hops labelled as collisions). A rule that is linear in these three features therefore does not use all the information in the residual; whether a nonlinear rule on the same features would close the gap was not tested.
- In S1–S3a terms, a jammed hop at the S1 outage point has $D = 10.7\,\mathrm{dB}$ when the jammer uses its worst-case band fraction and $D = 0.5\,\mathrm{dB}$ when it spreads over $\rho = 0.1$. At 10.7 dB a jammed hop is almost always told apart from a fade or a collision of the same strength; at 0.5 dB it is confused with a collision in about 40 % of hops.

6 Limits

- The data come from the model of Section 1, so the classifiers are tested on the distribution they were trained on. A jammer type or a fade profile that is absent from training is not tested.
- The fade is a single gain drop per hop that the receiver does not track. A receiver that estimates the gain from the pilots would remove this signature, and the fade would then look like a lower SNR.
- The receiver knows σ^2 and the pilots, as in S3a; all hops are independent; the collision is a single interferer with the same modulation and timing.
- The three causes are compared at equal D, a choice made to remove the mean energy as an explanation, not a property of real channels.

7 Correspondence with the code

Mathematics	Code
four classes, Section 1	`generate()`
T, C, K	`features()`
$y_i = x_i s_i/\sigma$, sorted	`dataset()`
three classifiers	`train()`, `energy_inputs()`
best accuracy from T	`bayes_accuracy_energy()`
means and distribution of T, C, K	`VERIFICATION 1`
energy bound	`VERIFICATION 2`
chi-square ROC bound	`VERIFICATION 3`
$a = 1 + D$ at S3a operating points	`noise_ratio()` from S3a, S1 link budget

References

[1] F. Pedregosa *et al.*, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.